

01 Assignment - Jenkins

Name: Pooja Prakash Belgaumkar

1. L1 – Create Jenkins Master-Slave Configurations and Create a workspace in Jenkins Slave Node using Jenkins Free-style project.

- Launch new EC2 instance for Slave (Agent).
- Allow SSH (22) from Jenkins Master's IP.
- Use same key pair as Master.

 [EC2](#) > [Instances](#) > Launch an instance   

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Jenkins Instance 2

[Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



De




Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

- Login to Slave Node and install Java.

```
ec2-user@ip-172-31-14-214:~  
POOJA@LAPTOP-LDU9LR85 MINGW64 ~/Downloads  
$ ssh -i linux1.pem ec2-user@13.62.57.229  
The authenticity of host '13.62.57.229 (13.62.57.229)' can't be established.  
ED25519 key fingerprint is SHA256:EWj8LITJKtqR9Gmf8n3s9K+6pd9oo1LFF97QQY/2TKI.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '13.62.57.229' (ED25519) to the list of known hosts.  
  
#_      Amazon Linux 2023  
~\     #####  
~\    #####  
~ \   ###|  
~  \#/  
~   V~'->  
~..-..  
~/_/'  
~/m/'  
  
[ec2-user@ip-172-31-14-214 ~]$ sudo yum install java-17-amazon-corretto -y  
Amazon Linux 2023 Kernel Livepatch repository    228 kB/s | 30 kB    00:00  
Dependencies resolved.  
  
=====
```

Package	Arch	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.17+10-1.amzn2023.1	amazonlinux	219 k
Installing dependencies:				
alsa-lib	x86_64	1.2.7.2-1.amzn2023.0.2	amazonlinux	504 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k

- Create Jenkins user on Slave.
- And create workspace directory.

```
jenkins@ip-172-31-14-214:~  
libXtst-1.2.5-1.amzn2023.0.1.x86_64  
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64  
libpng-2:1.6.37-10.amzn2023.0.8.x86_64  
libxcb-1.17.0-1.amzn2023.0.1.x86_64  
pixman-0.43.4-1.amzn2023.0.4.x86_64  
xml-common-0.6.3-56.amzn2023.0.2.noarch  
  
Complete!  
[ec2-user@ip-172-31-14-214 ~]$ java -version  
openjdk version "17.0.17" 2025-10-21 LTS  
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)  
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixed mode,  
sharing)  
[ec2-user@ip-172-31-14-214 ~]$ sudo useradd jenkins  
[ec2-user@ip-172-31-14-214 ~]$ sudo passwd jenkins  
Changing password for user jenkins.  
New password:  
BAD PASSWORD: The password contains the user name in some form  
Retype new password:  
Sorry, passwords do not match.  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[ec2-user@ip-172-31-14-214 ~]$ su - jenkins  
Password:  
[jenkins@ip-172-31-14-214 ~]$ mkdir -p /home/jenkins/jenkins_workspace  
[jenkins@ip-172-31-14-214 ~]$
```

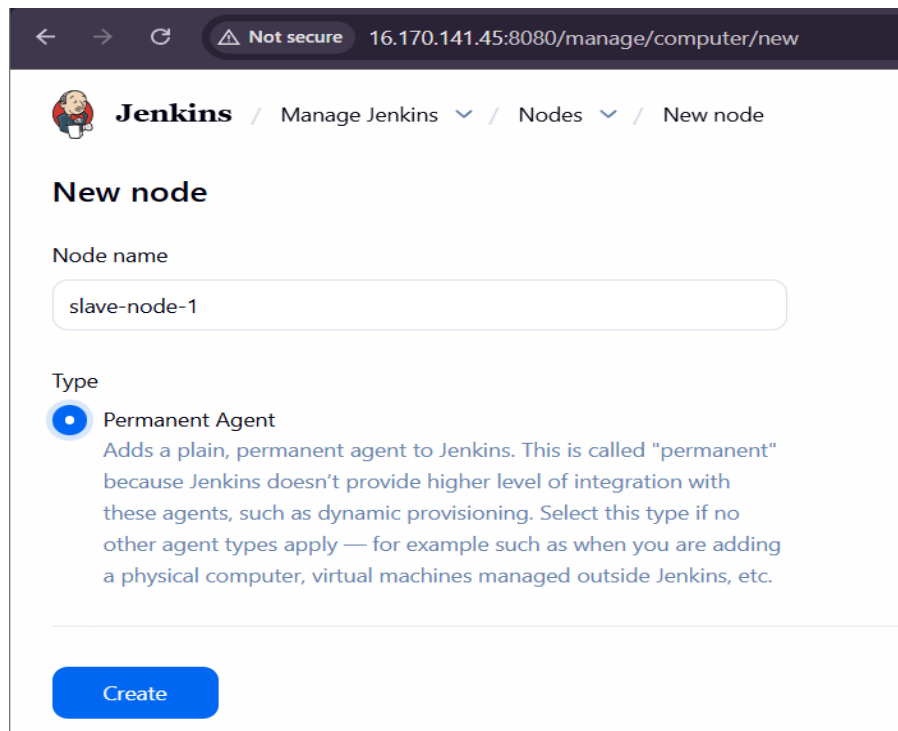
- Setup SSH Access on Master EC2, switch to Jenkins user and generate SSH key.

```
ec2-user@ip-172-31-7-48:~
Jan 19 09:09:11 ip-172-31-7-48.eu-north-1.compute.internal jenkins[1435]: 2026->
Jan 19 09:09:11 ip-172-31-7-48.eu-north-1.compute.internal jenkins[1435]: 2026->
Jan 19 09:09:52 ip-172-31-7-48.eu-north-1.compute.internal jenkins[1435]: 2026->
[ec2-user@ip-172-31-7-48 ~]$ sudo su - jenkins
[ec2-user@ip-172-31-7-48 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:XCgYlguebTjsf+SB2WsEHicrHoltlg738c1gAKmbG5M ec2-user@ip-172-31-7-48.eu-no
rth-1.compute.internal
The key's randomart image is:
+---[RSA 3072]-----+
|      .o.      |
| . =..o   .    |
| * B...  . .   |
|o * B  o  .    |
|o*.= B   S     |
|EoB = B        |
|.X.o B  *      |
|... o * o      |
|      o        |
+---[SHA256]-----+
[ec2-user@ip-172-31-7-48 ~]$
```

- Copy public key to Slave, if login works SSH setup is done.

[illegible]

- Login into the Jenkins through Master's EC2 and Create Slave node.



← → ↺ ⚠ Not secure 16.170.141.45:8080/manage/computer/new

 **Jenkins** / Manage Jenkins ▾ / Nodes ▾ / New node

New node

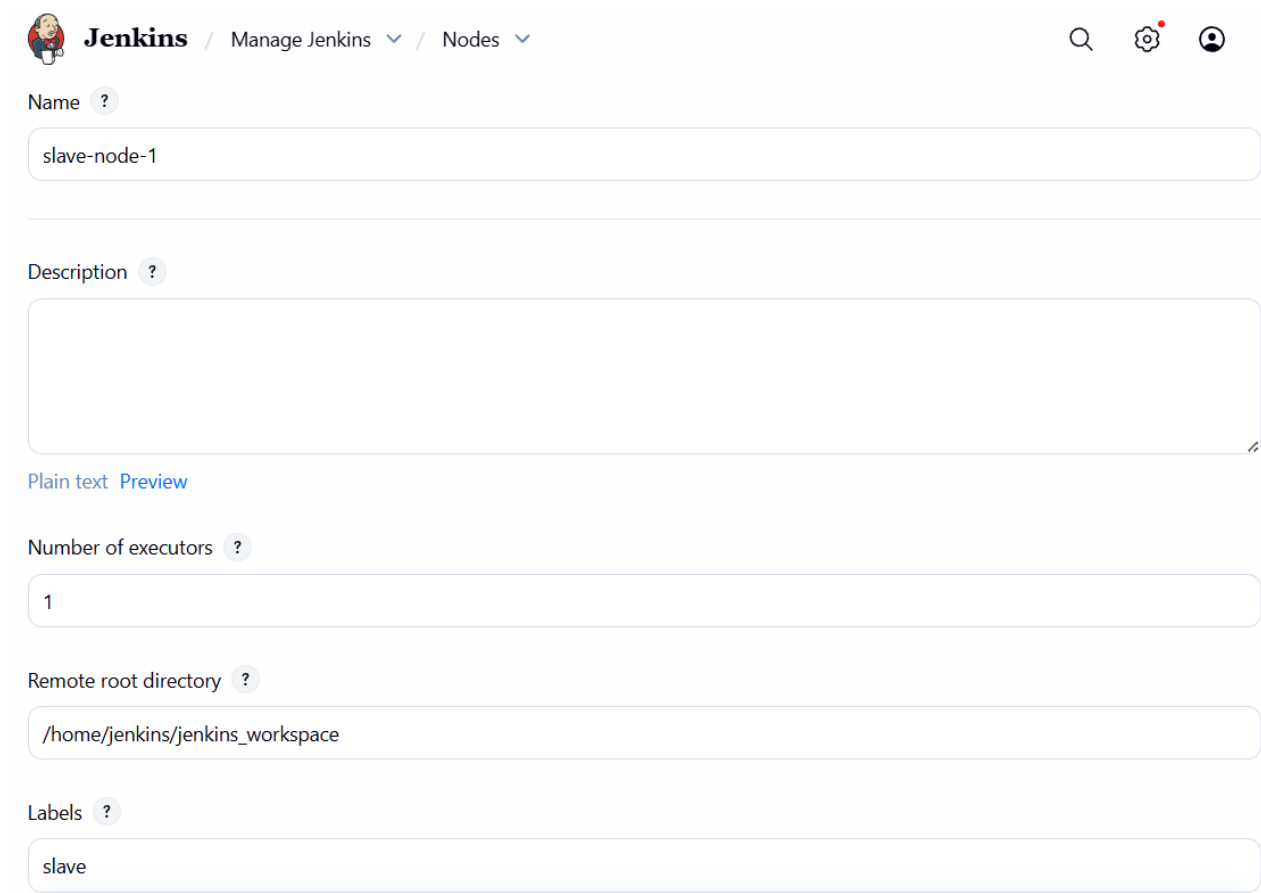
Node name

Type

☒ **Permanent Agent**

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create



 **Jenkins** / Manage Jenkins ▾ / Nodes ▾ 🔍 ⚙️ 👤

Name ?

Description ?

Plain text [Preview](#)

Number of executors ?

Remote root directory ?

Labels ?

 **Jenkins** / Manage Jenkins ▾ / Nodes ▾

🔍 ⚙️ 👤

Usage ?

Only build jobs with label expressions matching this node ▾

Launch method ?

Launch agents via SSH ▾

Host ?

Slave private IP

Credentials ?

jenkins ▾

+ Add

Host Key Verification Strategy ?

Non verifying Verification Strategy ▾

Advanced ▾

Availability ?

Keep this agent online as much as possible ▾

Save

- Create Free-Style project named Freestyle_proj_1.

 **Jenkins** / All ▾ / New Item

🔍 ⚙️ 👤

New Item

Enter an item name

Freestyle_proj_1

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

slave

Label slave matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

≡ **Execute shell** ?



Command

See [the list of available environment variables](#)

```
echo "This job is running on slave node"
hostname
pwd
```

Advanced ▾

+ Add build step

Save

Apply

- Then build and verify the workspace.

The screenshot shows the Jenkins web interface. At the top, the breadcrumb navigation reads 'Jenkins / Freestyle_proj_1 / #1 / Console Output'. On the left sidebar, there are links for 'Status', 'Changes', 'Console Output' (which is selected and highlighted), 'Edit Build Information', 'Delete build '#1'', and 'Timings'. The main area is titled 'Console Output' with a green checkmark icon. Above the output text are three buttons: 'Download', 'Copy', and 'View as plain text'. The console output text is as follows:

```
Started by user Pooja Prakash Belgaumkar
Running as SYSTEM
Building remotely on slave-node-1 (slave) in workspace
/home/jenkins/workspace/Freestyle_proj_1
[Freestyle_proj_1] $ /bin/sh -xe
/tmp/jenkins8121589637584421701.sh
+ echo 'This job is running on slave node'
This job is running on slave node
+ hostname
ip-172-31-14-214.eu-north-1.compute.internal
+ pwd
/home/jenkins/workspace/Freestyle_proj_1
Finished: SUCCESS
```

- Jenkins Master handles job scheduling and configuration, while Jenkins Slave executes the build jobs.
- The workspace is created on the slave node where the job runs, improving scalability and performance.